Core Capabilities at a Glance

- High-pressure boiler pressure parts: water wall panels, primary/secondary/radiant superheaters, economisers, bed & in-bed coils, steam headers, reboilers
- Specialty alloy fabrication in SS 304, SS 310, SS 316, and IBR/ASME-grade carbon & alloy steels
- Piping systems: HP/MP/LP steam lines, process piping, PRDS stations, IBR-certified pipe fittings
- Field & shutdown services: erection, commissioning, scaffolding, mechanical overhauling, NDT, thickness mapping, hydraulic testing, and reconditioning of pressure vessels & drums
- Engineering & QA/QC: ASME quality engineering, Authorised Inspector coordination, weld map preparation, MDR documentation, structural/lifting-lug design checks
- Manpower deployment: senior mechanical & E&I technicians for turbine maintenance and plant shutdowns
- Boiler retrofits & conversions, including multi-fuel firing upgrades (coal, bagasse, rice husk, RDF)
- Demonstrated international project delivery in the Gulf region (Qatar, Bahrain)

Industry Experience

The following sections summarise supply history and field-service work by sector.

1 Sugar, Distilleries, Bioenergy & Co-Generation

- High-pressure water wall panels (LHS, RHS, Front, Rear, Roof) with headers for cogeneration boilers, 15–50 TPH
- Economiser coils (BS 3059 Part-1 Gr 320 ERW) and bank tubes, OD 76.2 mm & 63.5 mm
- LP steam piping and Pressure Reducing & Desuperheating Station (PRDS) fabrication and site erection
- Complete boiler revival — dismantling, erection, and commissioning
- Bank tubes, platform grills, and superheater coils
- Pressure parts, bed coils, and process steam piping for distilleries and green-energy plants
- Cast-iron air seal castings for travelling grate furnaces (BHEL-make)
- Start-up and safety-valve vent silencers with acoustic packing for export sugar projects
- ID fan impellers (Sailma 350 HI) and spool-piece dummy plugs (SA 213 T22)
- Reconditioning, NDT, thickness mapping, and hydraulic testing of steam drums, water walls, and economisers for export projects

2 Pulp & Paper

- Multi-type bed coils (SA 210 Gr A1) with SS 304 studs; SS 304 secondary superheater coil bend sleeves
- Major boiler retrofits, including a 30 TPH AFBC to 20 TPH RC grate boiler conversion to fire 100% RDF alongside coal, bagasse, and rice husk
- Salt and ash screw conveyors
- HP/MP/LP steam headers (SA 335 P11, SA 106 Gr B, SA 515 Gr 60)
- Support pins and washers for economiser inlet headers; bottom economiser coil assemblies
- On-site pressure part erection, APH tube replacement, ducting modifications, and IBR compliance work
- Chemical recovery boiler capacity upgrades, e.g. 105 TPH to 120 TPH

3 Cement & Building Materials

- Front water wall tubes near IBSH coil boiler openings and evaporator locking arrangements
- Shutdown overhauls covering superheater assemblies, screen tubes, and ESP zone alignment
- Economiser/evaporator coil sets, boiler bank tubes, and ESP field erection
- Waste Heat Recovery Systems (WHRS) and AQC waste-heat boilers
- Bare carbon steel tubes (SA 210 Gr A), generator tubes with header coils, and spiral fin tube coil assemblies
- SS 304 scraper blades for limestone reclaimers; SS 304 water wall studs with ceramic ferrules
- Technical supervision for softener and water-treatment commissioning

4 Chemicals, Agro-Chemicals & Petrochemicals

- IBR-approved in-bed coils (SA 210 Gr A1 / SA 213) with TP347 studs, ceramic ferrules, and plate sleeves for coal-fired process boilers
- SS 304 SML buckets
- Shutdown technical services: scaffolding erection and mechanical overhauling
- SS 304 process tubes, IBR-approved piping spares, and specialty flow assemblies
- SS 304 fluidising air nozzles; studded in-bed/bed coil tubes, Types 1 & 2 (SA 210 Gr A1)
- Seamless carbon steel pipes with IBR test certificates

Clients include: *Privi Speciality Chemicals Limited, Reva Process Technologies*

5 Steel, Metallurgy & Heavy Industrial Plants

- Precision SS 310 fuel nozzle cap plates ("coal caps")
- Support tubes (top & bottom) with sleeves, loose tube bends, and SS 310 superheater support assemblies
- High-altitude process gas line replacement (1050NB) and thermowell replacement at 40–65 m
- Year-long O&M/AMC contracts for gas-dedusting (GDS) systems and coke oven batteries
- AFBC distributor plates and WHRB boiler headers
- Finned integral economiser assemblies; primary/secondary superheater coils with SS 304 supports (SA 213 T11/T22)
- Captive power plants and waste heat recovery systems within metallurgical complexes

6 Textiles, Yarn, Fabrics & Synthetic Fibers

- IBR-certified in-bed and bed coils (SA 210 Gr A1 / BS 3059 Gr 360) with SS 304 studs
- Custom water wall side panel membrane tubes with welded IS 2062 mild-steel fins
- High-temperature SS 316/304 support clamps for radiant superheaters
- Saturated shell-and-smoke-tube package steam boilers
- High-pressure IBR pipe fittings, fuel mixing nozzles, and steam line accessories
- Steam Coil Air Pre-Heater (SCAPH) erection and commissioning supervision
- Front, rear, and side water wall panels with MS fins (BS 3059 ERW)

7 Tyre & Rubber Manufacturing

- Extreme-temperature SS 310 bed coil sleeves (3", 600 mm)
- Fabricated bed coil tubes, Types C1 & C3 (SA 210 Gr A1 / BS 3059)
- Bedcoil assemblies (Cheema make) and in-bed coil sets (Thermax drawing)

- Economiser coil sets (BS 3059 ERW) and high-temperature SS 310 shield plates

Clients include: *Balkrishna Industries Limited (BKT), ATC Tires (Yokohama Off-Highway Tires)*

8 Food, Starch, Beverage, Dairy & Agro-Processing

- Custom IBR bed coils (Types I, II, III) with SS 304 studs; IBR-approved BS 3059 ST 360 bed coils
- Senior mechanical and E&I manpower deployment for turbine maintenance and plant shutdowns
- Process steam coils, in-bed coils, bank tubes, and pressure vessels
- Economiser assemblies, air distribution plates and nozzles
- Soot blower platforms, handrails, gratings, columns, and IBR-certified steam piping
- SS 304 sleeves with double-clamp plates for outer bed coils
- Convection/radiant superheater U-bends and thermowells/gauges

9 Power Generation & Captive Utilities

- Bed coils, water walls, and boiler-to-turbine steam line piping
- Primary and secondary superheater coil fabrication
- Boiler dust ducts, dust breakers, and end caps
- Superheater bends/tubes, evaporator and economiser tubes
- Combustor panels
- ASME quality engineering, Authorised Inspector coordination, weld map preparation, and MDR documentation
- Design strength checks, stiffener design, and lifting-lug verification for stack assemblies

10 Boiler OEMs, EPCs & Engineering Firms

- Water wall assemblies, primary/secondary convection superheaters, and radiant superheaters, built to custom drawings
- Custom studs and attemperator nozzles
- Reboilers (SA 213 TP316L)
- IBR steam headers (SA 335 P11)

11 Oil, Gas & International Infrastructure

- Specialised high-pressure IBR welding and site fabrication
- Technical engineering services, including stack-strength and lifting-lug design analysis

Clients include: *Gulf Asia Contracting (North Field Expansion Project), Valve Tech Testing & Services (Bahrain)*

Mining Sector — Ghana Opportunity

4.1 Ghana Mining Landscape — Snapshot (2026)

Proven Capability	Mining / Processing Application	Ghana-Specific Opportunity
Water walls, superheaters & economisers (captive power)	Captive / off-grid power generation at mine sites	Power infrastructure for Ashanti & Western goldfield mines and the VALCO smelter modernisation
Waste Heat Recovery Systems (WHRS)	Heat recovery from calciners, smelters & roasters	GIADEC alumina refinery & VALCO smelter; manganese beneficiation / ferroalloy plants
Pressure vessel NDT, testing & reconditioning	Pressure vessel & autoclave integrity management	Shutdown/turnaround support as new bauxite, manganese & lithium plants come online
Screw conveyors & ducting systems	Ore, ash & tailings handling	Material handling for the Ewoyaa DMS plant and Nsuta beneficiation
HP/MP/LP steam & process piping	Utility & process piping for refining circuits	Piping packages for gold processing plants and the planned alumina refinery
SS 304/310/316 high-temperature fabrication	High-wear kiln, dryer & roaster components	Bauxite calcination and manganese roasting/smelting equipment
Shutdown & turnaround field services	Planned outage support & manpower deployment	Field-service support for mine-site processing plants
ASME/IBR engineering & QA documentation	Documentation for heavy fabrication packages	Packages required by investors/EPCs entering GIADEC and Ewoyaa